# Schottky Barrier Rectifier, Trench-based, High Performance

# NRTS6100PFS, NRVTS6100PFS

This Trench Schottky rectifier is a high performance device in a TO-277 package. The lower forward voltage, less leakage current, and small junction capacitance are suitable to high switching frequency high density DC to DC conversion applications. It offers higher avalanche energy capability for Oring or reverse protection applications. The TO-277 package provides excellent thermal performance, less land area of board space, and a low profile.

### Features

- Lower Forward Voltage Drop
- Less Leakage Current in High Temperature
- Small Junction Capacitance for High Switching Frequency
- Higher Avalanche Energy Capability
- 175 °C Operating Junction Temperature
- Package Provided Capability of Inspection and Probe After Board Mounting
- Good Alternative Solution of SMC and DPAK Package
- NRV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable

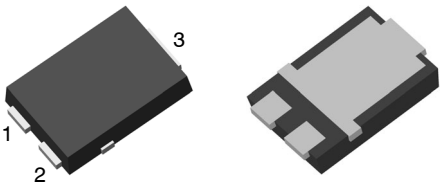
### Mechanical Characteristics:

- Case: Epoxy, Molded
- Epoxy Meets Flammability Rating UL 94-0 @ 0.125 in.
- Lead Finish: 100% Matte Sn (Tin)
- Lead and Mounting Surface Temperature for Soldering Purposes: 260 °C Max. for 10 Seconds
- Device Meets MSL 1 Requirements
- ESD Ratings:
  - Human Body Model: ≥ 8000 V (Class 3B)
  - Charged Device Model: > 1000 V (Class C5)

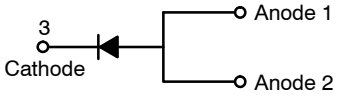
### Applications

- High Switching Frequency DC/DC Converters
- 2nd Rectifier
- Oring / Reverse Protection
- Freewheeling Diode for Inductive Loads

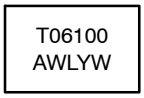
## SCHOTTKY BARRIER RECTIFIER, 6 AMPERES 100 VOLTS




**TO-277-3LD**
**CASE 340CZ**




### MARKING DIAGRAM




| | |
|---|---|
| T06100 | = Specific Device Code |
| A | = Assembly Location |
| Y | = Year |
| W | = Work Week |
| WL | = Wafer Lot |

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| NRTS6100PFST3G | TO-277 (Pb-Free) | 1500 / Tape & Reel |
| NRVTS6100PFST3G | TO-277 (Pb-Free) | 1500 / Tape & Reel |

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

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## MAXIMUM RATINGS

| Rating | Symbol | Value | Unit |
|---|---|---|---|
| Peak Repetitive Reverse Voltage<br>Working Peak Reverse Voltage<br>DC Blocking Voltage | $V_{RRM}$<br>$V_{RWM}$<br>$V_R$ | 100 | V |
| Continuous Forward Current<br>($T_C$ = 168 °C, DC) | $I_{F(DC)}$ | 6 | A |
| Peak Repetitive Forward Current,<br>($T_C$ = 166 °C, Square Wave, Duty = 0.5) | $I_{FRM}$ | 12 | A |
| Non-Repetitive Avalanche Energy<br>($T_J$ = 25 °C) | $E_{AS}$ | 145 | mJ |
| Storage Temperature Range | $T_{stg}$ | −65 to +175 | °C |
| Operating Junction Temperature (Note 1) | $T_J$ | −55 to +175 | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. The heat generated must be less than the thermal conductivity from Junction-to-Ambient $dP_D/dT_J < 1/R_{\theta JA}$

## THERMAL CHARACTERISTICS

| Characteristic | Symbol | Max | Unit |
|---|---|---|---|
| Thermal Resistance, Junction-to-Ambient | $R_{\theta JA}$ | 70 | °C/W |
| Thermal Resistance, Junction-to-Case Bottom | $R_{\theta JCB}$ | 2.2 | °C/W |
| Thermal Characterization, Junction-to-Case Top | $\Psi_{JCT}$ | 1.1 | °C/W |
| Thermal Characterization, Junction-to-Lead of Cathode | $\Psi_{JLC}$ | 0.7 | °C/W |

NOTE: (Assumes 600 $mm^2$, 1 oz. copper bond pad on a FR4 board)

## ELECTRICAL CHARACTERISTICS

| Characteristic | Symbol | Typ | Max | Unit |
|---|---|---|---|---|
| Instantaneous Forward Voltage (Note 2)<br>($I_F$ = 3 A, $T_J$ = 25 °C)<br>($I_F$ = 3 A, $T_J$ = 125 °C)<br>($I_F$ = 6 A, $T_J$ = 25 °C)<br>($I_F$ = 6 A, $T_J$ = 125 °C) | $V_F$ | <br>0.52<br>0.47<br>0.62<br>0.56 | <br><br><br>0.68<br>0.62 | V |
| Instantaneous Reverse Current (Note 2)<br>($V_R$ = Rated DC Voltage, $T_J$ = 25 °C)<br>($V_R$ = Rated DC Voltage, $T_J$ = 125 °C) | $I_R$ | <br>6.2<br>3.3 | <br>50<br>15 | <br>μA<br>mA |
| Junction Capacitance<br>($V_R$ = 1 V, $T_J$ = 25 °C, 1 MHz) | $C_J$ | <br>827 | | pF |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

2. Pulse Test: Pulse Width = 300 μs, Duty Cycle ≤ 2.0%.



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## TYPICAL CHARACTERISTICS

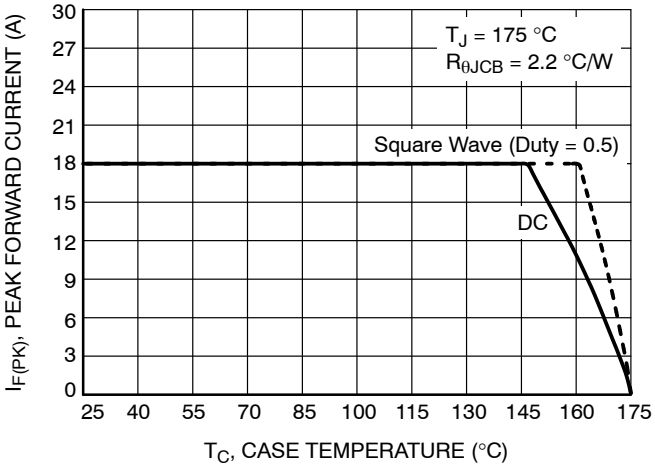



**Figure 1. Forward Current Derating of Case Temperature**

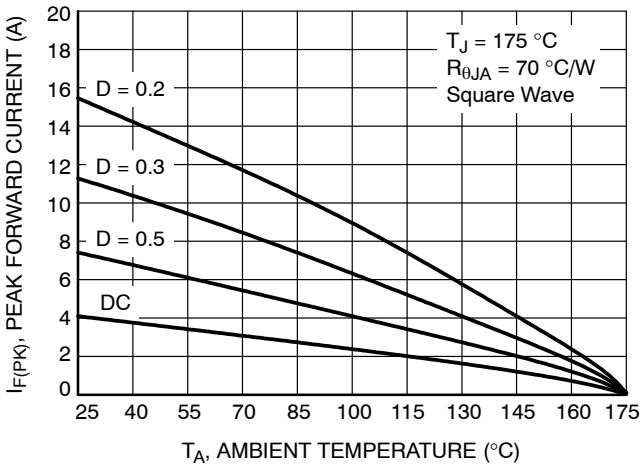



**Figure 2. Forward Current Derating of Ambient Temperature**

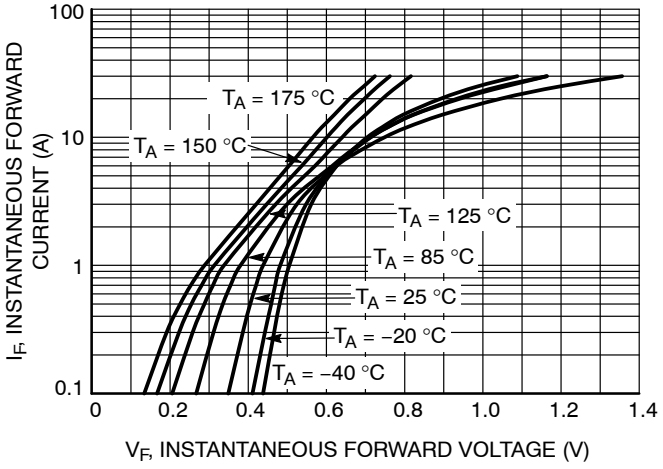



**Figure 3. Typical Forward Characteristics**

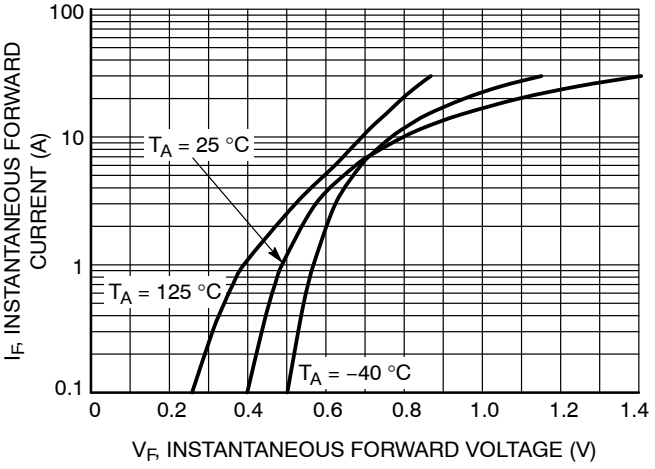



**Figure 4. Maximum Forward Characteristics**

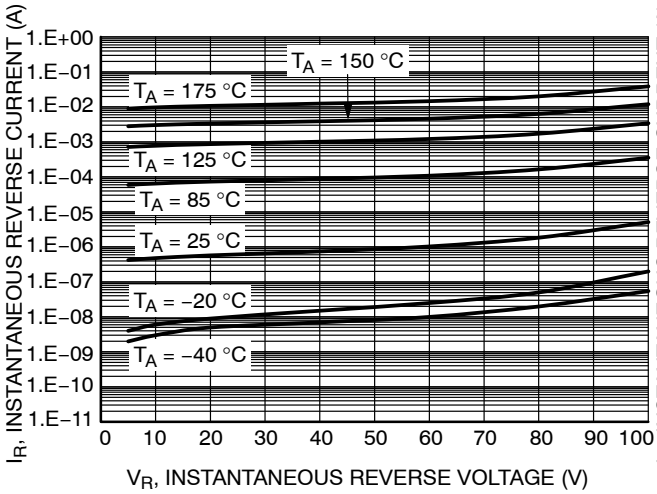



**Figure 5. Typical Reverse Characteristics**

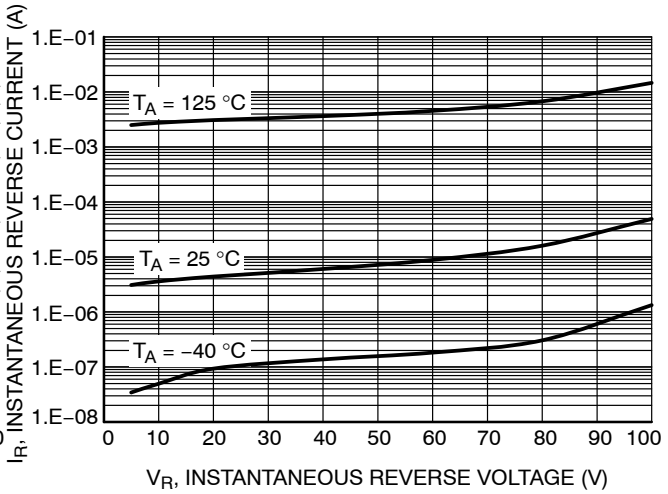



**Figure 6. Maximum Reverse Characteristics**



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## TYPICAL CHARACTERISTICS

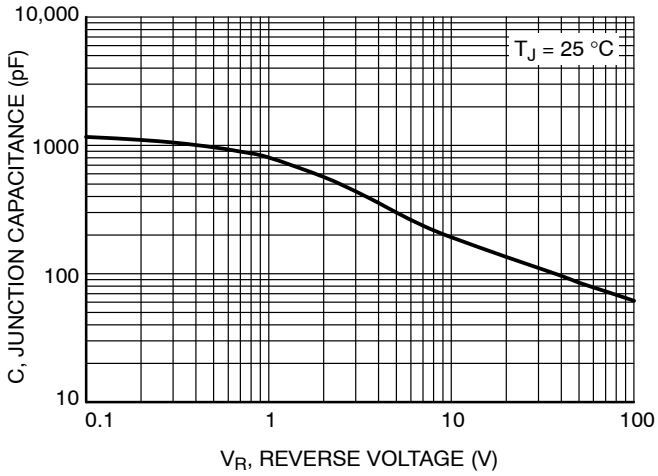



**Figure 7. Typical Junction Capacitance**

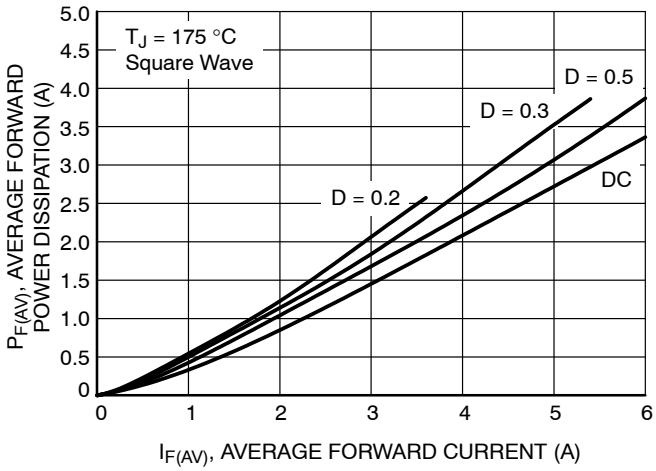



**Figure 8. Average Forward Power Dissipation**

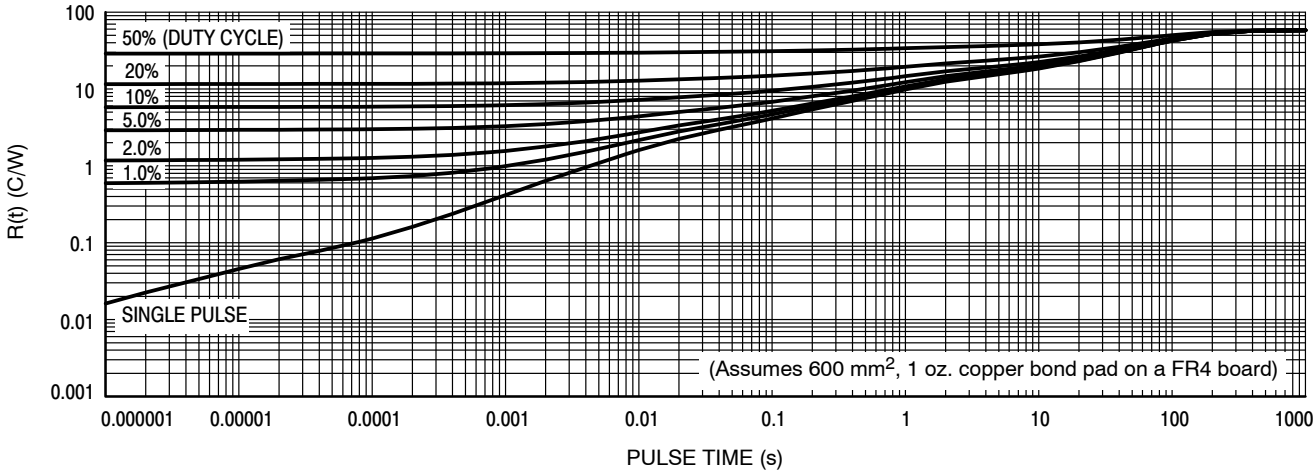



**Figure 9. Typical Thermal Characteristics, Junction-to-Ambient**



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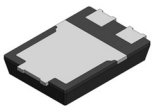
## REVISION HISTORY

| Revision | Description of Changes | Date |
|---|---|---|
| 2 | AEC-Q101 bullet added to the front page Features section. | 4/16/2026 |

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

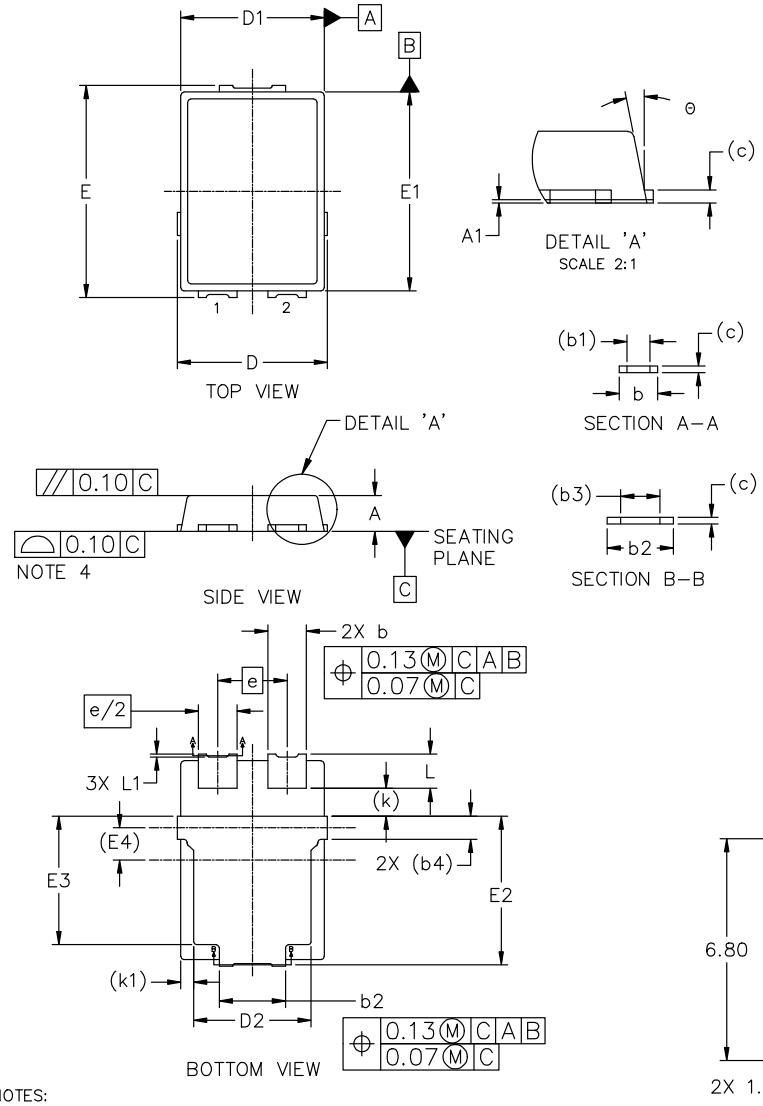


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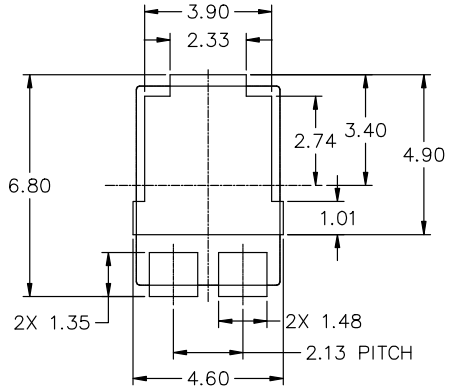


**TO−277−3 4.40x6.10x1.10, 2.13P**
CASE 340CZ
ISSUE B

DATE 04 MAY 2026




| MILLIMETERS | | | |
|---|---|---|---|
| DIM | MIN | NOM | MAX |
| A | 1.00 | 1.10 | 1.20 |
| A1 | −−− | 0.01 | 0.05 |
| b | 1.13 | 1.18 | 1.28 |
| b1 | | 0.70 REF | |
| b2 | 1.98 | 2.03 | 2.13 |
| b3 | | 1.20 REF | |
| b4 | | 0.71 REF | |
| c | | 0.20 REF | |
| D | 4.45 | 4.60 | 4.75 |
| D1 | 4.35 | 4.40 | 4.45 |
| D2 | 3.50 | 3.60 | 3.70 |
| E | 6.35 | 6.50 | 6.65 |
| E1 | 6.05 | 6.10 | 6.15 |
| E2 | 4.50 | 4.60 | 4.70 |
| E3 | 3.84 | 3.94 | 4.04 |
| E4 | | 0.98 REF | |
| e | | 2.13 BSC | |
| k | | 0.85 REF | |
| k1 | | 0.40 REF | |
| L | 0.90 | 1.05 | 1.20 |
| L1 | 0.02 | −−− | −−− |
| Θ | −−− | −−− | 12° |




RECOMMENDED MOUNTING FOOTPRINT

*For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERM/D.

NOTES:
1. DIMENSIONING AND TOLERANCING AS PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. DIMENSION b, b1, b2, b3, b4 and c TO BE MEASURED ON FLAT SECTION OF THE LEAD, BETWEEN 0.13 AND 0.25mm FROM LEAD TIP.
4. COPLANARITY APPLIES TO THE EXPOSED PAD AS WELL AS THE TERMINALS.
5. POSITIONAL TOLERANCE APPLIES TO THE TERMINALS AND EXPOSED PAD.
6. A1 IS DEFINED AS THE VERTICAL DISTANCE FROM THE SEATING PLANE TO THE LOWEST POINT ON THE PACKAGE BODY.
7. DIMENSION D1 AND E1 TO BE DETERMINED AT DATUM PLANE C.

**GENERIC MARKING DIAGRAM***




XXXXXX = Specific Device Code
A = Assembly Location
Y = Year
W = Work Week
WL = Wafer Lot

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot " ▪", may or may not be present. Some products may not follow the Generic Marking.

| | | |
|---|---|---|
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